

23 Dimensionality Reduction

Dimensionality reduction is the process of taking data in a high dimensional space and mapping it into a new space whose dimensionality is much smaller. This process is closely related to the concept of (lossy) compression in information theory. There are several reasons to reduce the dimensionality of the data. First, high dimensional data impose computational challenges. Moreover, in some situations high dimensionality might lead to poor generalization abilities of the learning algorithm (for example, in Nearest Neighbor classifiers the sample complexity increases exponentially with the dimension—see Chapter 19). Finally, dimensionality reduction can be used for interpretability of the data, for finding meaningful structure of the data, and for illustration purposes.

In this chapter we describe popular methods for dimensionality reduction. In those methods, the reduction is performed by applying a linear transformation to the original data. That is, if the original data is in $\mathbb{R}^d$ and we want to embed it into $\mathbb{R}^n$ ($n < d$) then we would like to find a matrix $W \in \mathbb{R}^{n,d}$ that induces the mapping $\mathbf{x} \mapsto W\mathbf{x}$. A natural criterion for choosing W is in a way that will enable a reasonable recovery of the original $\mathbf{x}$. It is not hard to show that in general, exact recovery of $\mathbf{x}$ from $W\mathbf{x}$ is impossible (see Exercise 1).

The first method we describe is called Principal Component Analysis (PCA). In PCA, both the compression and the recovery are performed by linear transformations and the method finds the linear transformations for which the differences between the recovered vectors and the original vectors are minimal in the least squared sense.

Next, we describe dimensionality reduction using random matrices W . We derive an important lemma, often called the “Johnson-Lindenstrauss lemma,” which analyzes the distortion caused by such a random dimensionality reduction technique.

Last, we show how one can reduce the dimension of all *sparse* vectors using again a random matrix. This process is known as Compressed Sensing. In this case, the recovery process is nonlinear but can still be implemented efficiently using linear programming.

We conclude by underscoring the underlying “prior assumptions” behind PCA and compressed sensing, which can help us understand the merits and pitfalls of the two methods.

23.1 Principal Component Analysis (PCA)

Let $\mathbf{x}_1, \dots, \mathbf{x}_m$ be m vectors in $\mathbb{R}^d$. We would like to reduce the dimensionality of these vectors using a linear transformation. A matrix $W \in \mathbb{R}^{n,d}$, where $n < d$, induces a mapping $\mathbf{x} \mapsto W\mathbf{x}$, where $W\mathbf{x} \in \mathbb{R}^n$ is the lower dimensionality representation of $\mathbf{x}$. Then, a second matrix $U \in \mathbb{R}^{d,n}$ can be used to (approximately) recover each original vector $\mathbf{x}$ from its compressed version. That is, for a compressed vector $\mathbf{y} = W\mathbf{x}$, where $\mathbf{y}$ is in the low dimensional space $\mathbb{R}^n$, we can construct $\tilde{\mathbf{x}} = U\mathbf{y}$, so that $\tilde{\mathbf{x}}$ is the recovered version of $\mathbf{x}$ and resides in the original high dimensional space $\mathbb{R}^d$.

In PCA, we find the compression matrix W and the recovering matrix U so that the total squared distance between the original and recovered vectors is minimal; namely, we aim at solving the problem

$$\operatorname{argmin}_{W \in \mathbb{R}^{n,d}, U \in \mathbb{R}^{d,n}} \sum_{i=1}^m \|\mathbf{x}_i - UW\mathbf{x}_i\|_2^2. \quad (23.1)$$

To solve this problem we first show that the optimal solution takes a specific form.

LEMMA 23.1 *Let (U, W) be a solution to Equation (23.1). Then the columns of U are orthonormal (namely, $U^\top U$ is the identity matrix of $\mathbb{R}^n$) and $W = U^\top$.*

Proof Fix any U, W and consider the mapping $\mathbf{x} \mapsto UW\mathbf{x}$. The range of this mapping, $R = \{UW\mathbf{x} : \mathbf{x} \in \mathbb{R}^d\}$, is an n dimensional linear subspace of $\mathbb{R}^d$. Let $V \in \mathbb{R}^{d,n}$ be a matrix whose columns form an orthonormal basis of this subspace, namely, the range of V is R and $V^\top V = I$. Therefore, each vector in R can be written as $V\mathbf{y}$ where $\mathbf{y} \in \mathbb{R}^n$. For every $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathbb{R}^n$ we have

$$\|\mathbf{x} - V\mathbf{y}\|_2^2 = \|\mathbf{x}\|^2 + \mathbf{y}^\top V^\top V \mathbf{y} - 2\mathbf{y}^\top V^\top \mathbf{x} = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\mathbf{y}^\top (V^\top \mathbf{x}),$$

where we used the fact that $V^\top V$ is the identity matrix of $\mathbb{R}^n$. Minimizing the preceding expression with respect to $\mathbf{y}$ by comparing the gradient with respect to $\mathbf{y}$ to zero gives that $\mathbf{y} = V^\top \mathbf{x}$. Therefore, for each $\mathbf{x}$ we have that

$$VV^\top \mathbf{x} = \operatorname{argmin}_{\tilde{\mathbf{x}} \in R} \|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2.$$

In particular this holds for $\mathbf{x}_1, \dots, \mathbf{x}_m$ and therefore we can replace U, W by $V, V^\top$ and by that do not increase the objective

$$\sum_{i=1}^m \|\mathbf{x}_i - UW\mathbf{x}_i\|_2^2 \geq \sum_{i=1}^m \|\mathbf{x}_i - VV^\top \mathbf{x}_i\|_2^2.$$

Since this holds for every U, W the proof of the lemma follows. $\square$

On the basis of the preceding lemma, we can rewrite the optimization problem given in Equation (23.1) as follows:

$$\operatorname{argmin}_{U \in \mathbb{R}^{d,n} : U^\top U = I} \sum_{i=1}^m \|\mathbf{x}_i - UU^\top \mathbf{x}_i\|_2^2. \quad (23.2)$$

We further simplify the optimization problem by using the following elementary algebraic manipulations. For every $\mathbf{x} \in \mathbb{R}^d$ and a matrix $U \in \mathbb{R}^{d,n}$ such that $U^\top U = I$ we have

$$\begin{aligned}\|\mathbf{x} - UU^\top \mathbf{x}\|^2 &= \|\mathbf{x}\|^2 - 2\mathbf{x}^\top UU^\top \mathbf{x} + \mathbf{x}^\top UU^\top UU^\top \mathbf{x} \\ &= \|\mathbf{x}\|^2 - \mathbf{x}^\top UU^\top \mathbf{x} \\ &= \|\mathbf{x}\|^2 - \text{trace}(U^\top \mathbf{x} \mathbf{x}^\top U),\end{aligned}\tag{23.3}$$

where the trace of a matrix is the sum of its diagonal entries. Since the trace is a linear operator, this allows us to rewrite Equation (23.2) as follows:

$$\operatorname{argmax}_{U \in \mathbb{R}^{d,n}: U^\top U = I} \text{trace} \left(U^\top \sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^\top U \right).\tag{23.4}$$

Let $A = \sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^\top$. The matrix A is symmetric and therefore it can be written using its spectral decomposition as $A = VDV^\top$, where D is diagonal and $V^\top V = VV^\top = I$. Here, the elements on the diagonal of D are the eigenvalues of A and the columns of V are the corresponding eigenvectors. We assume without loss of generality that $D_{1,1} \geq D_{2,2} \geq \dots \geq D_{d,d}$. Since A is positive semidefinite it also holds that $D_{d,d} \geq 0$. We claim that the solution to Equation (23.4) is the matrix U whose columns are the n eigenvectors of A corresponding to the largest n eigenvalues.

THEOREM 23.2 *Let $\mathbf{x}_1, \dots, \mathbf{x}_m$ be arbitrary vectors in $\mathbb{R}^d$, let $A = \sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^\top$, and let $\mathbf{u}_1, \dots, \mathbf{u}_n$ be n eigenvectors of the matrix A corresponding to the largest n eigenvalues of A . Then, the solution to the PCA optimization problem given in Equation (23.1) is to set U to be the matrix whose columns are $\mathbf{u}_1, \dots, \mathbf{u}_n$ and to set $W = U^\top$.*

Proof Let $VDV^\top$ be the spectral decomposition of A . Fix some matrix $U \in \mathbb{R}^{d,n}$ with orthonormal columns and let $B = V^\top U$. Then, $VB = VV^\top U = U$. It follows that

$$U^\top AU = B^\top V^\top V D V^\top V B = B^\top D B,$$

and therefore

$$\text{trace}(U^\top AU) = \sum_{j=1}^d D_{j,j} \sum_{i=1}^n B_{j,i}^2.$$

Note that $B^\top B = U^\top VV^\top U = U^\top U = I$. Therefore, the columns of B are also orthonormal, which implies that $\sum_{j=1}^d \sum_{i=1}^n B_{j,i}^2 = n$. In addition, let $\tilde{B} \in \mathbb{R}^{d,d}$ be a matrix such that its first n columns are the columns of B and in addition $\tilde{B}^\top \tilde{B} = I$. Then, for every j we have $\sum_{i=1}^d \tilde{B}_{j,i}^2 = 1$, which implies that $\sum_{i=1}^n B_{j,i}^2 \leq 1$. It follows that:

$$\text{trace}(U^\top AU) \leq \max_{\beta \in [0,1]^d: \|\beta\|_1 \leq n} \sum_{j=1}^d D_{j,j} \beta_j.$$

It is not hard to verify (see Exercise 2) that the right-hand side equals to $\sum_{j=1}^n D_{j,j}$. We have therefore shown that for every matrix $U \in \mathbb{R}^{d,n}$ with orthonormal columns it holds that $\text{trace}(U^\top AU) \leq \sum_{j=1}^n D_{j,j}$. On the other hand, if we set U to be the matrix whose columns are the n leading eigenvectors of A we obtain that $\text{trace}(U^\top AU) = \sum_{j=1}^n D_{j,j}$, and this concludes our proof. $\square$

Remark 23.1 The proof of Theorem 23.2 also tells us that the value of the objective of Equation (23.4) is $\sum_{i=1}^n D_{i,i}$. Combining this with Equation (23.3) and noting that $\sum_{i=1}^m \|\mathbf{x}_i\|^2 = \text{trace}(A) = \sum_{i=1}^d D_{i,i}$ we obtain that the optimal objective value of Equation (23.1) is $\sum_{i=n+1}^d D_{i,i}$.

Remark 23.2 It is a common practice to “center” the examples before applying PCA. That is, we first calculate $\boldsymbol{\mu} = \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i$ and then apply PCA on the vectors $(\mathbf{x}_1 - \boldsymbol{\mu}), \dots, (\mathbf{x}_m - \boldsymbol{\mu})$. This is also related to the interpretation of PCA as variance maximization (see Exercise 4).

23.1.1 A More Efficient Solution for the Case $d \gg m$

In some situations the original dimensionality of the data is much larger than the number of examples m . The computational complexity of calculating the PCA solution as described previously is $O(d^3)$ (for calculating eigenvalues of A) plus $O(md^2)$ (for constructing the matrix A). We now show a simple trick that enables us to calculate the PCA solution more efficiently when $d \gg m$.

Recall that the matrix A is defined to be $\sum_{i=1}^m \mathbf{x}_i \mathbf{x}_i^\top$. It is convenient to rewrite $A = X^\top X$ where $X \in \mathbb{R}^{m,d}$ is a matrix whose i th row is $\mathbf{x}_i^\top$. Consider the matrix $B = XX^\top$. That is, $B \in \mathbb{R}^{m,m}$ is the matrix whose i, j element equals $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$. Suppose that $\mathbf{u}$ is an eigenvector of B : That is, $B\mathbf{u} = \lambda\mathbf{u}$ for some $\lambda \in \mathbb{R}$. Multiplying the equality by $X^\top$ and using the definition of B we obtain $X^\top XX^\top \mathbf{u} = \lambda X^\top \mathbf{u}$. But, using the definition of A , we get that $A(X^\top \mathbf{u}) = \lambda(X^\top \mathbf{u})$. Thus, $\frac{X^\top \mathbf{u}}{\|X^\top \mathbf{u}\|}$ is an eigenvector of A with eigenvalue of λ .

We can therefore calculate the PCA solution by calculating the eigenvalues of B instead of A . The complexity is $O(m^3)$ (for calculating eigenvalues of B) and $m^2 d$ (for constructing the matrix B).

Remark 23.3 The previous discussion also implies that to calculate the PCA solution we only need to know how to calculate inner products between vectors. This enables us to calculate PCA implicitly even when d is very large (or even infinite) using kernels, which yields the *kernel PCA* algorithm.

23.1.2 Implementation and Demonstration

A pseudocode of PCA is given in the following.